HACKATHON PROJECT

TEMPERATURE LOGGER CALIBRATION PROVENANCE

A trustworthy digital record of how, when, where and against what standard a temperature logger was calibrated.

Team name:
Suraj gaur
P.Pranay
B.Ruthwik
S.Santhosh

Concept proposal • Prototype-ready architecture

Project overview

One platform connects a logger, its calibration evidence and every later change.

The idea

Build a digital provenance system that gives every temperature logger a unique identity and verifiable calibration history. A QR code opens its record, while signed, time-stamped events preserve the evidence trail.

IDENTITY

Unique logger ID, make, model, serial number and owner.

EVIDENCE

Certificate, reference standard, readings and uncertainty.

INTEGRITY

Role-based approval, audit log and tamper-evident hashes.

ACCESS

QR lookup, status and calibration due date.

Outcome: faster audits, fewer missing certificates, clearer calibration status and better confidence in temperature data.

Problem statement

Calibration may happen in a lab, but its evidence is often scattered across paper and systems.

FRAGMENTED RECORDS

Certificates, spreadsheets, service notes and logger IDs are stored separately. Linking the right document to the right device takes time.

WEAK TRACEABILITY

It can be difficult to prove which reference standard, procedure, technician and environmental conditions were used.

LIMITED VISIBILITY

Users may not know whether a logger is in date, overdue, adjusted, repaired or awaiting approval.

Core problem

There is no single, easy-to-verify lifecycle record connecting the physical logger to its calibration method, reference standards, results, uncertainty, certificate and subsequent history.

Who faces this problem?

The platform supports the full chain: from calibration laboratory to equipment user.

CALIBRATION LAB

Records measurements, standards, environmental conditions, adjustments and approvals.

QUALITY MANAGER

Checks certificate validity, traceability evidence, audit history and overdue devices.

EQUIPMENT OWNER

Finds current status, certificate, next due date and service history.

AUDITOR / CUSTOMER

Verifies the record and reviews evidence without searching multiple folders.

Example settings: food and pharmaceutical cold-chain monitoring, laboratories, hospitals, warehouses, environmental monitoring and industrial process control.

Proposed solution

A device passport plus a controlled calibration-event ledger.

1. DIGITAL LOGGER PASSPORT

Permanent device profile: unique ID, manufacturer, model, serial number, range, resolution, location, owner and QR label.

2. CALIBRATION EVENT

Structured record: date, procedure, technician, reference standards, points, as-found/as-left results, uncertainty and certificate.

3. VERIFICATION & APPROVAL

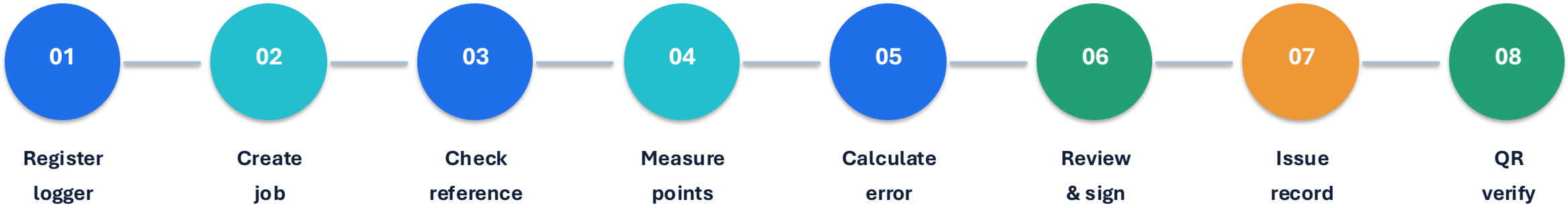
Reviewer checks completeness, validity of standards and acceptance criteria before marking the record approved.

4. TRACEABLE ACCESS

QR scan opens a read-only view of current status and approved history. Changes create new events, not silent overwrites.

End-to-end workflow

From device registration to certificate verification.



Control: a calibration event is not approved until required fields, reference-standard validity and reviewer sign-off are complete.

What data must be captured?

A strong provenance record explains both the result and how it was obtained.

LOGGER DETAILS

Unique asset ID • serial number
Make / model • sensor type
Range • resolution • location

CALIBRATION SETUP

Date and time • lab
Procedure / method revision
Technician • ambient conditions
Stabilisation time

REFERENCE EVIDENCE

Reference thermometer ID
Certificate number • due date
Calibration provider
Traceability statement

RESULTS

Set point
Reference value
Logger indication
Error / correction

UNCERTAINTY & DECISION

Uncertainty budget, coverage factor, acceptance tolerance and pass/fail decision with rationale.

DOCUMENTS & APPROVAL

Certificate PDF, raw data, attachments, reviewer, digital signature and event timestamp.

System architecture

A practical web application with secure storage and an append-only event history.

USER INTERFACE	Technician portal • Quality dashboard • QR verification page
APPLICATION API	Authentication • Device registry • Calibration workflow • Approval rules
PROVENANCE SERVICE	Event log • Hash generation • Digital signatures • Version control
DATA & FILE STORAGE	Relational database • Certificate/object storage • Backup
INTEGRATION LAYER	Logger CSV import • Reference-standard register • Optional sensor/IoT gateway

How provenance and integrity work

Make the history verifiable, not just editable.

APPEND-ONLY EVENTS

Create a new event for calibration, repair, adjustment or approval. Preserve the previous version and record who changed what and when.

HASH LINKING

Calculate a cryptographic hash for each event and link it to the previous event hash. A changed record can then be detected.

DIGITAL SIGN-OFF

Authenticate users and sign approved certificates or event summaries. Store signer identity, timestamp and signature evidence.

Security controls

- Role-based access: technician, reviewer, administrator and public verifier.
- Encryption in transit and at rest; secure backups and retention policy.
- QR page exposes only approved, non-sensitive information; private raw data remains access-controlled.

Calibration result: a simple example

The system stores raw observations and calculated values; it does not replace a validated procedure.

Illustrative measurement point

Field	Value	Unit
Reference temperature	5.00	°C
Logger indication	5.30	°C
Indication error	+0.30	°C

A full calibration uses multiple points across the required operating range, repeats where the method requires, and records as-found and as-left results.

FORMULA

Error = Logger indication − Reference value

Error = 5.30 − 5.00 = +0.30 °C

DECISION LOGIC

Compare result and uncertainty with the approved tolerance for intended use. Store the decision rule and reviewer approval.

How we will build the prototype

Start with a working minimum viable product; add automation after the evidence workflow works.

FRONT END

React web app

Login and role-based screens

Device profile and calibration form

QR verification page

BACK END

Python FastAPI / Node.js API

Validation and workflow rules

Hash and signature service

PDF certificate generation

DATA LAYER

PostgreSQL database

Object storage for certificates

Audit/event table

Scheduled due-date alerts

MVP build sequence

- Create logger registration, login and QR identity.
- Build calibration form, reference-standard register and result calculations.
- Add reviewer approval, immutable event history, certificate PDF and verification page.
- Test with sample data and one demonstration logger; document limitations.

Hackathon demo: follow one logger

A short story that judges can understand in under three minutes.

1	**Scan**	Scan QR label on logger LGR-001.
2	**Inspect**	Show make, model, range and current status.
3	**Calibrate**	Enter three sample points and reference-standard ID.
4	**Approve**	Reviewer checks results and signs the event.
5	**Verify**	Open QR page and show certificate, due date and event history.

Demo success: trace one device from identity → measurement evidence → approval → public verification.

Expected benefits and impact

Value is created through faster retrieval, clearer accountability and stronger evidence.

FASTER AUDITS

Find device history and certificates from one QR scan.

FEWER ERRORS

Required fields and validation reduce incomplete records.

BETTER CONTROL

Due-date alerts and status visibility support maintenance planning.

TRUSTED HISTORY

Signed events and hash checks help detect unauthorised changes.

Potential success metrics for pilot

- Time required to retrieve an approved certificate.
- Percentage of logger records with complete mandatory provenance fields.
- Number of overdue devices identified before use.
- Number of record-integrity verification checks passed.

Risks, validation and roadmap

Be clear about what the prototype proves—and what needs formal validation.

RISKS & MITIGATIONS

Incorrect input → validation and reviewer checks

QR copied → show device ID and verify physical label

Data loss → backups and recovery tests

False confidence → clear status and limitations

VALIDATION PLAN

Test role permissions and audit history

Verify hash mismatch detection

Check calculations against hand-worked examples

Test certificate download and QR lookup

Review security and backup recovery

Roadmap

HACKATHON

Clickable prototype + sample logger

PILOT

Use in one lab with controlled records

INTEGRATE

CSV imports, logger APIs and alerts

SCALE

Multi-site reporting and integrations

THE PROPOSAL

Make every temperature reading backed by a traceable story.

Temperature Logger Calibration Provenance turns scattered calibration documents into a connected, verifiable device history.

THANK YOU